

Progress Report: Ghost Lanes — Evaluating Lane Injection Attacks on Autonomous Driving Systems

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I. COMPLETED TASKS

We have successfully established the experimental framework and completed the primary attack evaluation phase in the CARLA simulator. Key achievements include:

- **Model Integration and Calibration:** Successfully integrated SCNN [1] architecture within a custom CARLA-Rosetta bridge. We calibrated the synthetic search window to a 480x320 resolution, ensuring stable detection on Mac M1 silicon hardware. The original plan was to integrate Ultra-Fast Lane Detection [2] as well, but this did not work on the hardware used.
- **Attack Matrix Execution:** Completed systematic evaluations for three injection geometries: Parallel, Convergent, and Divergent. We verified that synthetic road decals successfully bias the steering PD controller, as discussed in previous physical attack literature [3].
- **Quantitative Metrics Collection:** Successfully generated JSON formatted metrics for all attack configurations, tracking Maximum Lateral Deviation (MLD) and Attack Success Rate (ASR).
- **Preliminary Defense Validation:** Implemented a modular defense script utilizing temporal consistency and geometric width filtering [4].

Ghost Lanes Attack Analysis

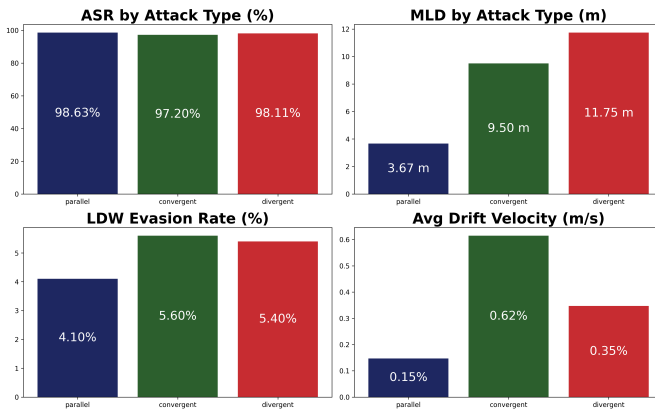


Fig. 1. Attack Analysis Plot

II. PLANNED TASKS (NEXT STEPS)

These tasks are scheduled for the final phase of the project:

- **Defense Parameter Optimization:** The current temporal threshold (2.0) and width range (100-200) resulted in a 13.6% rejection rate. We plan to implement a *Multi-Lane Confidence Consistency Check* to improve precision against smooth lateral drifts [5].
- **Cross-Model Transferability Study:** Evaluate if adversarial injection markings optimized for SCNN successfully transfer to the row-anchor-based Ultra-Fast model.
- **Environmental Robustness Testing:** Run selected attack configurations under varying weather conditions in CARLA Town03 to assess lighting-based degradation.
- **Defense Parameter Optimization:** The current temporal threshold (2.0) and width range (100-200) resulted in a 13.6% rejection rate. We plan to implement a *Multi-Lane Confidence Consistency Check* to improve precision against smooth lateral drifts [5].
- **Cross-Model Transferability Study:** Try again to get the Ultra-Fast Lane Detection to function properly and then evaluate if adversarial injection markings optimized for SCNN successfully transfer to the row-anchor-based Ultra-Fast model.
- **Environmental Robustness Testing:** Run selected attack configurations under varying weather conditions (Overcast, Dusk) in CARLA Town03 to assess lighting-based degradation.
- **Final Documentation:** Compile all visualizations into the final conference-style paper, including lateral deviation plots and defense precision/recall trade-off curves.

REFERENCES

- [1] P. L. X. W. X. Pan, J. Shi and X. Tang, "Spatial as deep: Spatial CNN for traffic scene understanding," in *Proc. AAAI Conf. on Artificial Intelligence*, 2018.
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- [3] T. S. et al., "Dirty road signs in the rearview mirror are closer than they appear: Reversible and reproducible physical-world attacks on autonomous vehicle steering," in *Proc. 30th USENIX Security Symposium*, 2021.
- [4] D. W. et al., "A systematic survey of adversarial attacks on deep learning based lane detection systems," *IEEE Transactions on Intelligent Transportation Systems*, 2024.
- [5] L. C. et al., "Revisiting adversarial perception attacks and defense methods on autonomous driving systems," *arXiv preprint*, 2505.